2-6-7

1. Let *f* be the function defined by $f(x) = k\sqrt{x} - \ln x,\ for\, x > 0$, where *k* is a positive constant. Find *f'(x)* and *f''(x)*.

Part A

1 point is earned for: *f'(x)*

1 point is earned for: *f''(x)*

$$f\prime(x) = \frac{k}{2\sqrt{x}} = -\frac{1}{x}$$
$$f\prime\prime(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

0	1	2

The student response earns all of the following points:

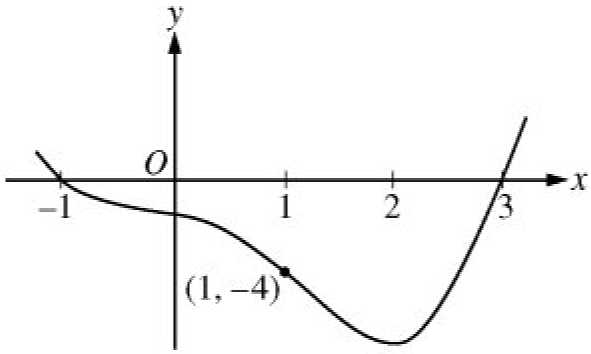
1 point is earned for: *f'(x)*

1 point is earned for: *f''(x)*

$$f\prime(x) = \frac{k}{2\sqrt{x}} = -\frac{1}{x}$$
$$f\prime\prime(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

2-6-7

2.



Graph of f'

Let f be a twice-differentiable function defined on the interval $-1.2<x<3.2$ with $f(1)=2$. The graph of f', the derivative of f, is shown above. The graph of f' crosses the x-axis at $x=-1$ and $x=3$ and has a horizontal tangent at $x=2$. Let g be the function given by $g(x)=e^{f(x)}$.

Write an equation for the line tangent to the graph of g at $x=1$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$g(1) = e^{f(1)} = e^2$

$g'(x) = e^{f(x)} f'(x),\ g'(1)=e^{f(1)} f'(1) = -4e^2$

The tangent line is given by $y = e^2 - 4e^2(x-1)$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$g(1) = e^{f(1)} = e^2$

$g'(x) = e^{f(x)} f'(x),\ g'(1)=e^{f(1)} f'(1) = -4e^2$

The tangent line is given by $y = e^2 - 4e^2(x-1)$.

2-6-7

3. Let f be the function given by the $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1 - \ln x}{x^2}$.

The graph of the function f has exactly one point of inflection. Find the x-coordinate of this point.

Part C

The response can earn up to 3 points:

2 point: For $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2\ln x}{x^3} \text{ for all } x > 0 f''(x) = 0 \text{ when } -3 + 2\ln x = 0 x = e^{3/2}$$

1 point: For the correct answer

The graph of f has a point of inflection at x = e3/2 because $f''(x)$ changes sign at x = e3/2

0	1	2	3

The response earns all three of the following points:

2 point: For $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2\ln x}{x^3} \text{ for all } x > 0 f''(x) = 0 \text{ when } -3 + 2\ln x = 0 x = e^{3/2}$$

1 point: For the correct answer

The graph of f has a point of inflection at x = e3/2 because $f''(x)$ changes sign at x = e3/2

2-6-7

4. Let f be the function given by $f(x) = \ln|x/1 + x^2|$.

(a) Find the domain of f.

(b) Determine whether f is an even function, an odd function, or neither. Justify your conclusion.

(c) At what values of x does f have a relative maximum or a relative minimum? For each such x, use the first derivative test to determine whether $f(x)$ is a relative maximum or a relative minimum.

(d) Find the range of f.

Part A

1 point is earned for answer

$x \neq 0$

0	1

The student response earns all of the following points:

1 point is earned for answer

$x \neq 0$

Part B

1 point is earned for the answer

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln\left|\frac{-x}{1 + (-x)^2}\right| = f(x)$$

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

2-6-7

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln\left|\frac{-x}{1+(-x)^2}\right| = f(x)$$

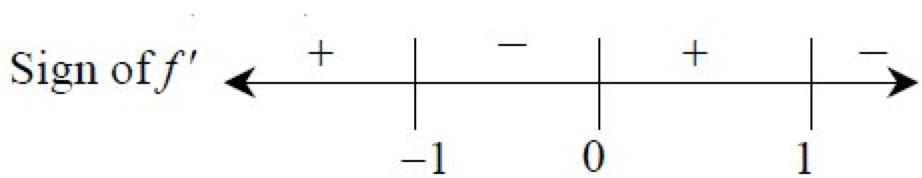
Part C

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$\begin{aligned} f'(x) &= \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2} \\ &= \frac{1-x^2}{x(1+x^2)} \end{aligned}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$

0	1	2	3	4

The student response earns all of the following points:

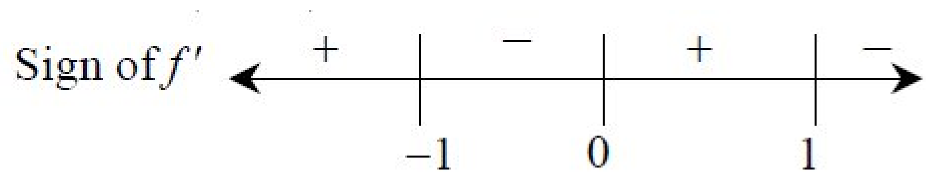
2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

2-6-7

$$f'(x) = \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$
$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at x = 1

$f(x)$ has relative max at x = -1

Part D

2 points are earned for the answer

1 point **deducted** for error

max is

$$f(1) = \ln\frac{1}{2} = -\ln 2$$
$$\lim_{x \to 0^+} f(x) = -\infty$$
$$\left(\text{or} \lim_{x \to 0^-} f(x) = \lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = -\infty\right)$$
$$\therefore \text{range} = \left(-\infty, \ln\frac{1}{2}\right]$$

0	1	2

The student response earns all of the following points:

2 points are earned for the answer

1 point **deducted** for error

max is

2-6-7

$$f(1) = \ln \frac{1}{2} = -\ln 2$$
$$\lim_{x \to 0^{+}} f(x) = -\infty$$
$$\left(\text{or} \lim_{x \to 0^{-}} f(x) = \lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = -\infty\right)$$
$$\therefore \text{range} = \left(-\infty, \ln \frac{1}{2}\right]$$

2-6-7

5. Let f be the function given by $f(x)=\ln|x/1+x^2|$

At what values of x does f have a relative maximum or a relative minimum? For each such x, use the first derivative test to determine whether $f(x)$ is a relative maximum or a relative minimum.

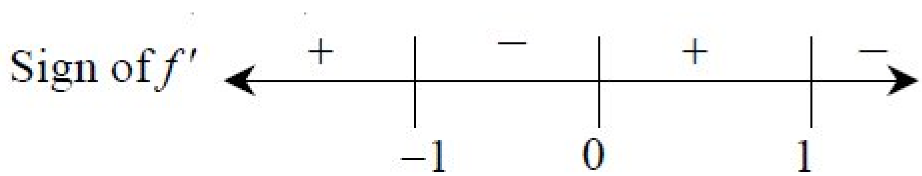
Part C

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of f'(x) and conclusion

$$f'(-x)=\frac{1-x^2}{x}.\frac{(1+x^2)-2x^2}{(1+x^2)^2}$$
$$=\frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at $x=1$
$f(x)$ has relative max at $x=-1$

0	1	2	3	4

The student response earns four of the following points:

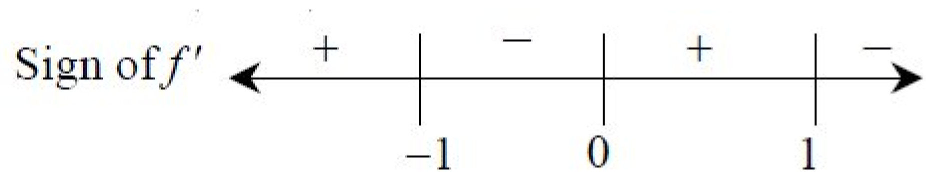
2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of f'(x) and conclusion

$$f'(-x)=\frac{1-x^2}{x}.\frac{(1+x^2)-2x^2}{(1+x^2)^2}$$
$$=\frac{1-x^2}{x(1+x^2)}$$

2-6-7



$f(x)$ has relative max at $x = 1$
$f(x)$ has relative max at $x = -1$

2-6-7

6. Let f be a function defined by $f(x) = \begin{cases} 1 - 2\sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

(a) Show that f is continuous at $x = 0$.

(b) For $x \neq 0$, express $f'(x)$ as a piecewise-defined function. Find the value of x for which $f'(x) = -3$.

(c) Find the average value of f on the interval [-1, 1].

Part A

2 point is earned for analysis

$\lim_{x \to 0^-} (1 - 2\sin x) = 1$

$\lim_{x \to 0^+} e^{-4x} = 1$

$f(0) = 1$

So, $\lim_{x \to 0} f(x) = f(0)$.

Therefore f is continuous at $x = 0$.

0	1	2

The student response earns all of the following points:

2 point is earned for analysis

$\lim_{x \to 0^-} (1 - 2\sin x) = 1$

$\lim_{x \to 0^+} e^{-4x} = 1$

$f(0) = 1$

So, $\lim_{x \to 0} f(x) = f(0)$.

Therefore f is continuous at $x = 0$.

2-6-7

Part B

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2\cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2\cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4}\text{In}\left(\frac{3}{4}\right) > 0.$$

Therefore, $f'(x) = -3$ for $x = -\frac{1}{4}\text{In}\left(\frac{3}{4}\right)$.

0	1	2	3

The student response earns all of the following points:

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2\cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2\cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4}\text{In}\left(\frac{3}{4}\right) > 0.$$

2-6-7

Therefore, $f'(x) = -3$ for $x = -\frac{1}{4}\text{In}\left(\frac{3}{4}\right)$.

Part C

1 point is earned for $\int_{-1}^{0}(1 - 2\sin x)dx$ and $\int_{0}^{1} e^{-4x}dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned}\int_{-1}^{1} f(x)dx &= \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx \\ &= \int_{-1}^{0}(1 - 2\sin x)dx + \int_{0}^{1} e^{-4x}dx \\ &= [x + 2\cos]_{x=1}^{x=0} + \left[-\tfrac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2\cos\ (-1)) + \left(-\tfrac{1}{4}e^{-4} + \tfrac{1}{4}\right)\end{aligned}$$

$$\begin{aligned}\text{Average value} &= \tfrac{1}{2}\int_{-1}^{1} f(x)dx \\ &= \frac{13}{8} - \cos(-1) - \tfrac{1}{8}e^{-4}\end{aligned}$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $\int_{-1}^{0}(1 - 2\sin x)dx$ and $\int_{0}^{1} e^{-4x}dx$

2 point is earned for antiderivatives

1 point is earned for answer

2-6-7

$$\int_{-1}^{1} f(x)dx = \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx$$
$$= \int_{-1}^{0} (1 - 2\sin x)dx + \int_{0}^{1} e^{-4x}dx$$
$$= [x + 2\cos]_{x=1}^{x=0} + \left[-\tfrac{1}{4}e^{-4x}\right]_{x=0}^{x=1}$$
$$= (3 - 2\cos\ (-1)) + \left(-\tfrac{1}{4}e^{-4} + \tfrac{1}{4}\right)$$

$$\text{Average value} = \tfrac{1}{2}\int_{-1}^{1} f(x)dx$$
$$= \frac{13}{8} - \cos(-1) - \tfrac{1}{8}e^{-4}$$

7. Let f be the function defined by $f(x) = ln(x^2 + 1)$, and let g be the function defined by $g(x) = x^5 + x^3$. The line tangent to the graph of f at $x = 2$ is parallel to the line tangent to the graph of g at $x = a$, where a is a positive constant.

What is the value of a ?

(A) 0.246

(B) 0.430

(C) 0.447

(D) 0.790

8. What is the slope of the line tangent to the graph of $y = \frac{e^{-x}}{x+1}$ at $x = 1$?

(A) $-\frac{1}{e}$

(B) $-\frac{3}{4e}$

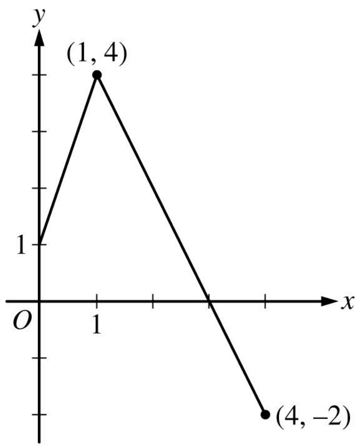
(C) $-\frac{1}{4e}$

(D) $\frac{1}{4e}$

(E) $\frac{1}{e}$

2-6-7

9.



Graph of f

The graph of the function f, consisting of two line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + 1$, and let h be the function given by $h(x) = f(g(x))$. What is the value of $h'(1)$?

(A) -4

(B) -2

(C) 4

(D) 6

(E) nonexistent

2-6-7

10. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

What is the acceleration of the object time $t = 4$?

Part A

1 point is earned for the answer

$$a\left(4\right) = v'\left(4\right) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6}\ or - 0.523\ or - 0.524$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$a\left(4\right) = v'\left(4\right) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6}\ or - 0.523\ or - 0.524$$

2-6-7

11. A particle moves along the x-axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$. Find the acceleration $a(t)$ of the particle at any time t.

Part A

2 points are earned for correct differentiation of velocity

$$a(t) = v\prime(t)$$
$$= -2\pi\cos(2\pi t)$$

0	1	2

The student response earns all of the following points:

2 points are earned for correct differentiation of velocity

$$a(t) = v\prime(T)$$
$$= 2\pi\cos(2\pi t)$$

2-6-7

12. Particle X moves along the positive x-axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$

Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$x'(t) = 15t^2 - 18t$

$x'(1) = 15 - 18 = -3$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$

0	1	2

The response earns both of the following points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$x'(t) = 15t^2 - 18t$

$x'(1) = 15 - 18 = -3$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

2-6-7

13. A particle moves along the x-axis with position at time t given by $x(t) = e^{-t}\sin t$ for $0 \leq t \leq 2\pi$. Find the time t at which the particle is farthest to the left. Justify your answer.

Part A

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$x\prime(t) = -e^{-t}\sin t + e^{-t}\cos t = -e^{-t}(\cos t - \sin t)$
$x\prime(t) = 0, when, \cos t = \sin t. Therefore, x\prime(t) = 0, on$
$0 \leq t \leq 2\pi, for, t = \frac{\pi}{4} and, t = \frac{5\pi}{4}$

The candidates for the absolute minimum are at

$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} and, 2\pi.$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

0	1	2	3	4	5

The student response earns all of the following points:

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

2-6-7

1 point is earned for the answer

1 point is earned for the justification

$x\prime(t) = -e^{-t}\sin t + e^{-t}\cos t = -e^{-t}(\cos t - \sin t)$
$x\prime(t) = 0, when, \cos t = \sin t. Therefore, x\prime(t) = 0, on$
$0 \leq t \leq 2\pi, for, t = \frac{\pi}{4} and, t = \frac{5\pi}{4}$

The candidates for the absolute minimum are at

$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} and, 2\pi.$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

2-6-7

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

For $0 \leq t \leq 6$ seconds, a screen saver on a computer screen shows two circles that start as dots and expand outward.

(a) At the instant that the first circle has a radius of 9 centimeters, the radius is increasing at a rate of $\frac{3}{2}$ centimeters per second. Find the rate at which the area of the circle is changing at that instant. Indicate units of measure.

(b) The radius of the first circle is modeled by $w(t) = 12 - 12e^{-0.5t}$ for $0 \leq t \leq 6$, where $w(t)$ is measured in centimeters and *t* is measured in seconds. At what time *t* is the radius of the circle increasing at a rate of 3 centimeters per second?

(c) A model for the radius of the second circle is given by the function *f* for $0 \leq t \leq 6$, where $f(t)$ is measured in centimeters and *t* is measured in seconds. The rate of change of the radius of the second circle is given by $f'(t) = t^2 - 4t + 4$. Based on this model, by how many centimeters does the radius of the second circle increase from time $t = 0$ to $t = 3$?

Part A

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

$$A = \pi r^2$$
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$
$$\left.\frac{dA}{dt}\right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi \text{ cm}^2/\text{sec}$.

✓

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

2-6-7

$A = \pi r^2$
$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$
$\left.\frac{dA}{dt}\right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi \text{ cm}^2/\text{ sec}$.

Part B

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$
$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2\ln 2$

The radius is increasing at a rate of 3 centimeters per second at time$t = 2\ln 2$seconds.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$
$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2\ln 2$

The radius is increasing at a rate of 3 centimeters per second at time $t = 2\ln 2$ seconds.

Part C

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

2-6-7

$$\int_0^3 \left(t^2 - 4t + 4\right)\,dt = \left[\tfrac{1}{3}t^3 - 2t^2 + 4t\right]_{t=0}^{t=3}$$
$$= \left(\tfrac{1}{3}\cdot 3^3 - 2\cdot 3^2 + 4\cdot 3\right) - \left(\tfrac{1}{3}\cdot 0^3 - 2\cdot 0^2 + 4\cdot 0\right)$$
$$= 9 - 18 + 12 = 3$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$\int_0^3 \left(t^2 - 4t + 4\right)\,dt = \left[\tfrac{1}{3}t^3 - 2t^2 + 4t\right]_{t=0}^{t=3}$$
$$= \left(\tfrac{1}{3}\cdot 3^3 - 2\cdot 3^2 + 4\cdot 3\right) - \left(\tfrac{1}{3}\cdot 0^3 - 2\cdot 0^2 + 4\cdot 0\right)$$
$$= 9 - 18 + 12 = 3$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

15. If $y = \sin^3 x$, then $\dfrac{dy}{dx} =$

(A) $\cos^3 x$

(B) $3\cos^2 x$

(C) $3\sin^2 x$

(D) $-3\sin^2 x\cos x$

(E) $3\sin^2 x\cos x$ ✓

16. If $f(x) = \cos^3(4x)$, then $f\prime(x) =$

(A) $3\cos^2(4x)$

(B) $-12\cos^2(4x)\sin(4x)$ ✓

(C) $-3\cos^2(4x)\sin(4x)$

(D) $12\cos^2(4x)\sin(4x)$

(E) $-4\sin^3(4x)$

2-6-7

17. Let *f* be the function given by $f(x) = \cos(2x) + \ln(3x)$. What is the least value of *x* at which the graph of *f* changes concavity?

(A) 0.56

(B) 0.93 ✓

(C) 1.18

(D) 2.38

(E) 2.44

18. If $\arcsin x = \ln y$, then $\frac{dy}{dx} =$

(A) $\frac{y}{\sqrt{1-x^2}}$ ✓

(B) $\frac{xy}{\sqrt{1-x^2}}$

(C) $\frac{y}{1+x^2}$

(D) $e^{\arcsin x}$

(E) $\frac{e^{\arcsin x}}{1+x^2}$

19. $\frac{d}{dx}(2^x)=$

(A) 2^{x-1}

(B) $\left(2^{x-1}\right)x$

(C) $\left(2^x\right)\ln 2$ ✓

(D) $\left(2^{x-1}\right)\ln 2$

(E) 2x/ln2

20. If $f(x) = \mathrm{e}^x$, then $\ln(f'(2)) =$

(A) 2 ✓

(B) 0

(C) $1/\mathrm{e}^2$

(D) 2e

(E) e^2

2-6-7

21. If $f(x) = ae^{-ax}$ for $a > 0$, then $f'(x) =$

(A) e^{-ax}

(B) ae^{-ax}

(C) a^2e^{-ax}

(D) $-ae^{-ax}$

(E) $-a^2e^{-ax}$ ✓

22. If $f(x) = \ln x$, then $\lim\limits_{x \to 3} \frac{f(x) - f(3)}{x - 3}$ is

(A) $\frac{1}{3}$ ✓

(B) e^3

(C) ln 3

(D) nonexistent

2-6-7

23. Let f be the function given by $f(x)=\ln(x / x-1)$

Find the value of the derivative of f at $x=-1$.

Part B

2 point is earned for finding the derivative

2 point is deducted for error in chain,product, quotient or derivative of ln.

1 point is deducted for all other error

1 point is earned for evaluating his/her f'(x) at $x=-1$

$f'(x)=\frac{x-1}{x}\cdot\frac{(x-1)-x}{(x-1)^2}$

$=\frac{-1}{x(x-1)}$

or

$\ln|x|-\ln|x-1| \Rightarrow f'(x)=\frac{1}{x}-\frac{1}{x-1}$

$f'(-1)=-\frac{1}{2}$

0	1	2	3

The student response earns three of the following points:

2 point is earned for finding the derivative

2 point is deducted for error in chain,product, quotient or derivative of ln.

1 point is deducted for all other error

1 point is earned for evaluating his/her f'(x) at $x=-1$

$f'(x)=\frac{x-1}{x}\cdot\frac{(x-1)-x}{(x-1)^2}$

$=\frac{-1}{x(x-1)}$

or

$\ln|x|-\ln|x-1| \Rightarrow f'(x)=\frac{1}{x}-\frac{1}{x-1}$

$f'(-1)=-\frac{1}{2}$

2-6-7

24. If $y = \frac{\ln x}{x}$, then $\frac{dy}{dx} =$

(A) $\frac{1}{x}$

(B) $\frac{1}{x^2}$

(C) $\frac{\ln x - 1}{x^2}$

(D) $\frac{1 - \ln x}{x^2}$

(E) $\frac{1 + \ln x}{x^2}$

25. Let f be the function given by $f(x) = \frac{\backslash\text{abs}x - 2}{x - 2}$.

Find $f'(1)$.

Part B

1 point is earned for $f'(0)$

$for\ x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f\prime(1) = 0$

0	1

The student response earns one of the following points:

1 point is earned for $f'(0)$

$for\ x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f\prime(1) = 0$

2-6-7

26. If $f(x) = \sin x$, then f′(π/3) =

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$ ✓

(C) $\frac{\sqrt{2}}{2}$

(D) $\frac{\sqrt{3}}{2}$

(E) $\sqrt{3}$

27. If $y = \sin(3x)$, then $\frac{dy}{dx} =$

(A) $-3\cos(3x)$

(B) $-\cos(3x)$

(C) $-\frac{1}{3}\cos(3x)$

(D) $\cos(3x)$

(E) $3\cos(3x)$ ✓

2-6-7

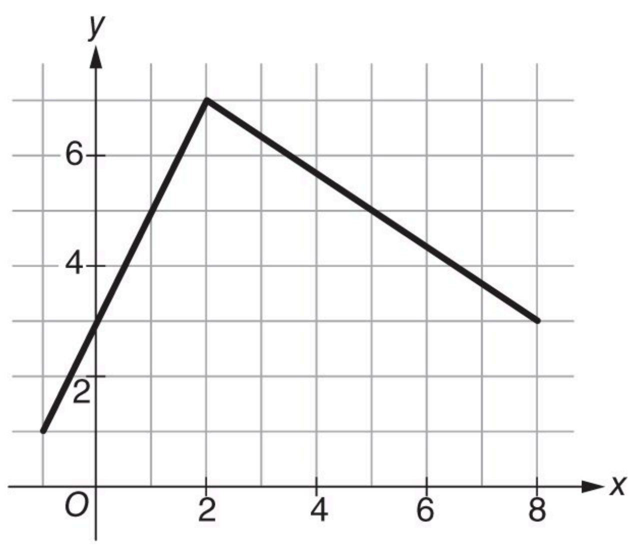
28. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

Let f be the continuous function defined on $[-1, 8]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = \sqrt{x^2 - x + 3}$ and $h(x) = 5e^x - 9\sin x$.

(a) The function k is defined by $k(x) = f(x)g(x)$. Find $k'(0)$.

(b) The function m is defined by $m(x) = \frac{f(x)}{2g(x)}$. Find $m'(5)$.

(c) Find the value of x for $-1 < x < 2$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(0)$ *or* $g'(0)$.

2-6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $f'(0)$
- [ ] $g'(0)$
- [ ] $k'(0)$

Solution:

$k'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$

$f'(0) = 2$ from the graph.

$g'(0) = -0.288675$

$k'(0) = f'(0) \cdot g(0) + f(0) \cdot g'(0)$
$= 2 \cdot \sqrt{3} + 3 \cdot (-0.288675) = 2.598$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $f'(5)$
- [ ] $g'(5)$
- [ ] $m'(5)$

Solution:

$$m'(x) = \frac{f'(x) \cdot 2g(x) - f(x) \cdot 2g'(x)}{(2g(x))^2}$$

2-6-7

$f'(5) = -\frac{2}{3}$ from the graph.

$g'(5) = 0.938315$

$$m'(5) = \frac{f'(5) \cdot 2g(5) - f(5) \cdot 2g'(5)}{(2g(5))^2}$$

$$= \frac{\left(-\frac{2}{3}\right) \cdot 2\sqrt{23} - 5 \cdot 2 \cdot 0.938315}{\left(2\sqrt{23}\right)^2} = -0.171$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$f'(x) = 2$ for $-1 < x < 2$

$h'(x) = 5e^x - 9\cos x$

$h'(x) = 2 \Rightarrow x = 0.622$

2-6-7

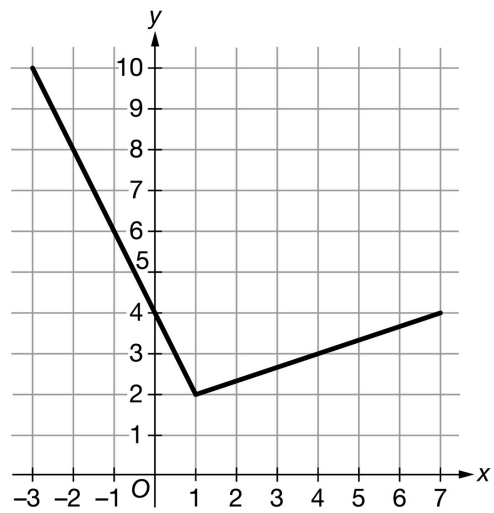
29. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

Let f be the continuous function defined on $[-3, 7]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = \left(x^2 + 5x\right)^{\frac{1}{3}}$ and $h(x) = 7\cos x + x^3$.

(a) The function N is defined by $N(x) = f(x)g(x)$. Find $N'(-1)$. Show the work that leads to your answer.

(b) The function P is defined by $P(x) = \frac{g(x)}{5f(x)}$. Find $P'(4)$. Show the work that leads to your answer.

(c) Find the value of x for $-3 < x < 1$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(-1)$ or $g'(-1)$.

2-6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ $f'(-1)$
- ☐ $g'(-1)$.
- ☐ $N'(-1)$

Solution:

$N'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$

$f'(-1) = -2$ determined from the slope of the line segment on the graph.

$g'(-1) = 0.396850$

$$N'(-1) = f'(-1) \cdot g(-1) + f(-1) \cdot g'(-1)$$
$$= (-2) \cdot (-4)^{\frac{1}{3}} + 6 \cdot 0.396850 = 5.556 \text{ (or } 5.555)$$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ $f'(4)$
- ☐ $g'(4)$
- ☐ $P'(4)$

Solution:

$$P'(x) = \frac{g'(x) \cdot 5f(x) - g(x) \cdot 5f'(x)}{(5f(x))^2}$$

2-6-7

$f'(4) = \frac{1}{3}$ determined from the slope of the line segment on the graph.

$g'(4) = 0.397454$

$$P'(4) = \frac{g'(4) \cdot 5f(4) - g(4) \cdot 5f'(4)}{(5f(4))^2}$$

$$= \frac{0.397454 \cdot 5 \cdot 3 - 36^{\frac{1}{3}} \cdot 5 \cdot \frac{1}{3}}{(5 \cdot 3)^2} = 0.002$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$f'(x) = -2$ determined from the slope of the line segment on the graph for $-3 < x < 1$

$h'(x) = -7\sin x + 3x^2$

$h'(x) = -2 \ \Rightarrow \ x = 0.343 \ (\text{or } 0.342)$

2-6-7

30. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$f(x)$	$f'(x)$
3	-1	8
5	4	3
7	6	-2

The table above gives values of the differentiable function f and its derivative f' at selected values of x. Let h be the function defined by $h(x) = x^2e^{\cos(3x)}$.

(a) Using values of f from the table, estimate $f'(4)$. Show the work that leads to your answer.

(b) Let M be the function defined by $M(x) = f(x) + h(x)$. Find $M'(3)$.

(c) Let N be the function defined by $N(x) = f(x)h(x)$. Find $N'(5)$.

(d) Let P be the function defined by $P(x) = \frac{h(x)}{f(x)}$. Find $P'(7)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-6-7

0	1

The student response accurately includes a correct estimate of $f'(4)$

Solution:

$$f'(4) \approx \frac{f(5)-f(3)}{5-3} = \frac{4-(-1)}{5-3} = \frac{5}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a sum, either by writing $M'(x) = f'(x) + h'(x)$ or $M'(3) = f'(3) + h'(3)$ or $M'(3) = 8 + (-2.061490)$ OR for the correct answer accompanied by $f'(3) = 8$ and $h'(3) = -2.061490$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $8 + (-2.061490)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $M'(x)$
- ☐ answer

Solution:

$M'(x) = f'(x) + h'(x)$

$f'(3) = 8$ from the table.

$h'(3) = -2.061490$ from the calculator.

$M'(3) = f'(3) + h'(3) = 5.939 \text{ (or } 5.938)$

Part C

2-6-7

The first point is earned for correctly presenting the product rule for derivatives, either by writing $N'(x) = f'(x)h(x) + f(x)h'(x)$ or $N'(5) = f'(5)h(5) + f(5)h'(5)$ or $N'(5) = 3(11.695310) + 4(-18.137830)$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $3(11.695310) + 4(-18.137830)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $N'(x)$
- [ ] answer

Solution:

$N'(x) = f'(x)h(x) + f(x)h'(x)$

$f(5) = 4$ and $f'(5) = 3$ from the table.

$h(5) = 11.695310$ and $h'(5) = -18.137830$ from the calculator.

$N'(5) = f'(5)h(5) + f(5)h'(5) = -37.465$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing $P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$ or $P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2}$ or $P'(7) = \frac{(-63.023773)\cdot 6 - 28.334809\cdot(-2)}{6^2}$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $\frac{(-63.023773)\cdot 6 - 28.334809\cdot(-2)}{6^2}$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-6-7

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $P'(x)$
- [ ] answer

Solution:

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$$

$f(7) = 6$ and $f'(7) = -2$ from the table.

$h(7) = 28.334809$ and $h'(7) = -63.023773$ from the calculator.

$$P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} = -8.930 \ \text{(or } -8.929\text{)}$$

2-6-7

31. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$g(x)$	$g'(x)$
2	-2	4
4	5	1
6	2	-3

The table above gives values of the differentiable function g and its derivative g' at selected values of x. Let f be the function defined by $f(x) = x \sin\left(\sqrt{3x^2+5}\right)$.

(a) Using values of g from the table, estimate $g'(5)$. Show the work that leads to your answer.

(b) Let R be the function defined by $R(x) = f(x) - g(x)$. Find $R'(2)$.

(c) Let S be the function defined by $S(x) = f(x)g(x)$. Find $S'(4)$.

(d) Let T be the function defined by $T(x) = \frac{f(x)}{g(x)}$. Find $T'(6)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-6-7

0	1

The student response accurately includes a correct estimate of $g'(5)$

Solution:

$$g'(5) \approx \frac{g(6)-g(4)}{6-4} = \frac{2-5}{6-4} = -\frac{3}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a difference, either by writing $R'(x) = f'(x) - g'(x)$ or $R'(2) = f'(2) - g'(2)$ or $R'(2) = -2.448854 - 4$ OR for the correct answer accompanied by $f'(2) = -2.448854$ and $g'(2) = 4$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $-2.448854 - 4$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $R'(x)$
- [ ] answer

Solution:

$R'(x) = f'(x) - g'(x)$

$f'(2) = -2.448854$ from the calculator.

$g'(2) = 4$ from the table.

$R'(2) = f'(2) - g'(2) = -6.449$

Part C

2-6-7

The first point is earned for correctly presenting the product rule for derivatives, either by writing $S'(x) = f'(x)g(x) + f(x)g'(x)$ or $S'(4) = f'(4)g(4) + f(4)g'(4)$ or

$S'(4) = (4.419230) \cdot (5) + (3.359221) \cdot (1)$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$(4.419230) \cdot (5) + (3.359221) \cdot (1)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] $S'(x)$
- [ ] answer

Solution:

$S'(x) = f'(x)g(x) + f(x)g'(x)$

$f(4) = 3.359221$ and $f'(4) = 4.419230$ from the calculator.

$g(4) = 5$ and $g'(4) = 1$ from the table.

$S'(4) = f'(4)g(4) + f(4)g'(4) = 25.455$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing $T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$ or $T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2}$ or

$T'(6) = \frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2}$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$\frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2}$ earns the second point.

2-6-7

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $T'(x)$
- [ ] answer

Solution:

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$f(6) = -5.603824$ and

$f'(6) = -4.564565$ from the calculator.

$g(6) = 2$ and $g'(6) = -3$ from the table.

$$T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} = -6.485$$

32. $\frac{d}{dx}\left(\sin^3\left(x^2\right)\right) =$

(A) $\cos^3\left(x^2\right)$

(B) $3\sin^2\left(x^2\right)$

(C) $6x\sin^2\left(x^2\right)$

(D) $3\sin^2\left(x^2\right)\cos\left(x^2\right)$

(E) $6x\sin^2\left(x^2\right)\cos\left(x^2\right)$ ✓

33. $\frac{d}{dx}$ ln \abscos $\left(\frac{\pi}{x}\right)$ is

(A) $\frac{-\pi}{x^2\cos\left(\frac{\pi}{x}\right)}$

(B) $-\tan\left(\frac{\pi}{x}\right)$

(C) $\frac{1}{\cos\left(\frac{\pi}{x}\right)}$

(D) $\frac{\pi}{x}\tan\left(\frac{\pi}{x}\right)$

(E) $\frac{\pi}{x^2}\tan\left(\frac{\pi}{x}\right)$ ✓

2-6-7

34. $d/dx(1/x^3 - 1/x + x^2)$ at $x = -1$ is

(A) -6

(B) -4 ✓

(C) 0

(D) 2

(E) 6

35. Suppose that *f* is an odd function; i.e., $f(-x)=-f(x)$ for all x. Suppose that $f'(x_0)$ exists. Which of the following must necessarily be equal to $f'(-x_0)$?

(A) $f'(x_0)$ ✓

(B) $-f'(x_0)$

(C) $\frac{1}{f'(x_0)}$

(D) $-\frac{1}{f'(x_0)}$

(E) None of the above

36. If $y = \frac{1}{2}x^{4/5} - \frac{3}{x^5}$, then $\frac{dy}{dx} =$

(A) $\frac{2}{5x^{1/5}} + \frac{15}{x^6}$ ✓

(B) $\frac{2}{5x^{1/5}} + \frac{15}{x^4}$

(C) $\frac{2}{5x^{1/5}} - \frac{3}{5x^4}$

(D) $\frac{2x^{1/5}}{5} + \frac{15}{x^6}$

(E) $\frac{2x^{1/5}}{5} - \frac{3}{5x^4}$

37. If $\ln(2x + y) = x + 1$, then $\frac{dy}{dx} =$

(A) -2

(B) $2x + y - 2$ ✓

(C) $2x + y$

(D) $4x + 2y - 2$

(E) $y - \frac{y}{x}$

2-6-7

38. What is the slope of the line tangent to the polar curve $r = 1 + 2\sin\theta$ at $\theta = 0$?

(A) 2

(B) $\frac{1}{2}$

(C) 0

(D) $-\frac{1}{2}$

(E) -2

2-6-7

39.

x	-2	$-2 < x < -1$	-1	$-1 < x < 1$	1	$1 < x < 3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x. Values of f, f', g, and g' for various values of x are given in the table above.

The function h is defined by $h(x)=\ln(f(x))$. Find $h'(3)$. Show the computations that lead to your answer.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

2-6-7

40. If $f(x) = 4x^{-2} + \frac{1}{4}x^2 + 4$, then $f'(2) =$

(A) -62

(B) -58

(C) -3

(D) 0

(E) 1

2-6-7

The twice-differentiable function f is defined for all real numbers and satisfies the following conditions: $f(0) = 2$, $f'(0) = -4$, and $f''(0) = 3$.

41. The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a. Show the work that leads to your answer.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$g'(x) = aeax + f'(x)$

$g'(0) = a - 4$

$g''(x) = a2eax + f''(x)$

$g''(0) = a2 + 3$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$g'(x) = aeax + f'(x)$

$g'(0) = a - 4$

$g''(x) = a2eax + f''(x)$

$g''(0) = a2 + 3$

2-6-7

42. The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part B

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$h'(x) = f'(x) \cos(kx) - k\sin(kx) f(x)$

$h'(0) = f'(0) \cos(0) - k\sin(0) f(0) = f'(0) = 4$

$h'(0) = \cos(0) f(0) = 2$

The equation of the tangent line is y = -4x + 2.

0	1	2	3	4	5

The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$h'(x) = f'(x) \cos(kx) - k\sin(kx) f(x)$

$h'(0) = f'(0) \cos(0) - k\sin(0) f(0) = f'(0) = 4$

$h'(0) = \cos(0) f(0) = 2$

The equation of the tangent line is y = -4x + 2.

2-6-7

43. If $y = x^2 e^x$, then $\frac{dy}{dx} =$

(A) $2xe^x$

(B) $x\left(x + 2e^x\right)$

(C) $xe^x\left(x + 2\right)$ ✓

(D) $2x + e^x$

(E) $2x + e$

44. The slope of the line tangent to the graph of $y = \ln\left(1 - x\right)$ at x = -1 is

(A) -1

(B) -1/2 ✓

(C) 1/2

(D) ln 2

(E) 1

45. If *f(x)=x* + sin *x*, then *f'(x)* =

(A) 1 + cos *x* ✓

(B) 1 - cos *x*

(C) cos *x*

(D) sin *x* - *x*cos *x*

(E) sin x + xcos x

46. If $f\left(x\right) = 7x - 3 + \ln\, x$, then $f'\left(1\right) =$

(A) 4

(B) 5

(C) 6

(D) 7

(E) 8 ✓

47. If $f\left(x\right) = \left(x - 1\right)^2 \sin x$, then $f'\left(0\right) =$

(A) -2

(B) -1

(C) 0

(D) 1 ✓

(E) 2

2-6-7

48. The slope of the line tangent to the graph of $y = \ln(1-x)$ at $x = -1$ is

(A) -1

(B) -1/2

(C) 1/2

(D) ln2

(E) 1

2-6-7

49. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions:

$$f(0) = 2, f'(0) = -4, \text{ and } f''(0) = 3.$$

(a) The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a. Show the work that leads to your answer.

(b) The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$
$$g'(0) = a - 4$$
$$g''(x) = a^2e^{ax} + f''(x)$$
$$g''(0) = a^2 + 3$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$
$$g'(0) = a - 4$$
$$g''(x) = a^2e^{ax} + f''(x)$$
$$g''(0) = a^2 + 3$$

Part B

2-6-7

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$h'(x) = f'(x)\cos(kx) - k\sin(kx)f(x)$

$h'(0) = f'(0)\cos(0) - k\sin(0)f(0) = f'(0) = -4$

$h(0) = \cos(0)f(0) = 2$

The equation of the tangent line is $y = -4x + 2$.

0	1	2	3	4	5 ✓

The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$h'(x) = f'(x)\cos(kx) - k\sin(kx)f(x)$

$h'(0) = f'(0)\cos(0) - k\sin(0)f(0) = f'(0) = -4$

$h(0) = \cos(0)f(0) = 2$

The equation of the tangent line is $y = -4x + 2$.

2-6-7

50. Let *f* be the function given by *f*(*x*) = ln(*x* / *x* − 1).

(a) *What is the domain of f ?*

(b) Find the value of the derivative of *f* at $x = -1$.

(c) Write an expression for $f^{-1}(x)$, *where* f^{-1} *denotes the inverse function of* f.

Part A

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$\frac{x}{x-1} > 0$

$x > 0$ and $x - 1 > 0 \Rightarrow x > 1$

$x < 0$ and $x - 1 < 0 \Rightarrow x < 0$

$\boxed{x < 0 \text{ or } x > 1}$

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$\frac{x}{x-1} > 0$

$x > 0$ and $x - 1 > 0 \Rightarrow x > 1$

$x < 0$ and $x - 1 < 0 \Rightarrow x < 0$

$\boxed{x < 0 \text{ or } x > 1}$

2-6-7

Part B

2 point is earned for finding the derivative

2 point is **deducted** for error in chain,product, quotient or derivative of ln.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f'(x)$ at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$

$$= \frac{-1}{x(x-1)}$$

or

$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$

$$\boxed{f'(-1) = -\frac{1}{2}}$$

0	1	2	3

The student response earns all of the following points:

2 point is earned for finding the derivative

2 point is **deducted** for error in chain,product, quotient or derivative of ln.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f'(x)$ at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$

$$= \frac{-1}{x(x-1)}$$

or

$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$

$$\boxed{f'(-1) = -\frac{1}{2}}$$

2-6-7

Part C

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$$y = \ln\left(\frac{x}{x-1}\right)$$
$$e^y = \frac{x}{x-1}$$
$$x(e^y - 1) = e^y$$
$$x = \frac{e^y}{e^y-1}$$
$$\boxed{f^{-1}(x) = \frac{e^x}{e^x-1}}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$$y = \ln\left(\frac{x}{x-1}\right)$$
$$e^y = \frac{x}{x-1}$$
$$x(e^y - 1) = e^y$$
$$x = \frac{e^y}{e^y-1}$$
$$\boxed{f^{-1}(x) = \frac{e^x}{e^x-1}}$$

2-6-7

51. $\lim_{h\to 0}\dfrac{e^{2+h}-e^2}{h}$ is

(A) 0

(B) e^2

(C) $2e^2$

(D) nonexistent

Answer B

Correct. The limit of this difference quotient is of the form $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$, where $f(x)=e^x$. This is one way to express the derivative of f at $x=2$. Since $f'(x)=e^x$, $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=f'(2)=e^2$.

52. $\lim_{h\to 0}\frac{e^{(2+h)}-e^2}{h}=$

(A) 0

(B) 1

(C) $2e$

(D) e^2

(E) $2e^2$

2-6-7

53. A particle moves along the x-axis with velocity given by $v(t) = \frac{10\sin(0.4t^2)}{t^2-t+3}$ for time $0 \le t \le 3.5$. The particle is at position $x = -5$ at time $t = 0$.

(a) Find the acceleration of the particle at time $t = 3$.

(b) Find the position of the particle at time $t = 3$.

(c) Evaluate $\int_0^{3.5} v(t)\,dt$, and evaluate $\int_0^{3.5} |v(t)|\,dt$. Interpret the meaning of each integral in the context of the problem.

(d) A second particle moves along the x-axis with position given by $x_2(t) = t^2 - t$ for $0 \le t \le 3.5$. At what time t are the two particles moving with the same velocity?

Part A

1 point is earned for answer

$v'(3) = -2.118$

The acceleration of the particle at time $t = 3$ is -2.118.

0	1

The student earns all of the following points:

1 point is earned for answer

$v'(3) = -2.118$

The acceleration of the particle at time $t = 3$ is -2.118.

Part B

1 point is earned for $\int_0^3 v(t)\,dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t)\,dt = -5 + \int_0^3 v(t)\,dt = -1.760213$$

2-6-7

The position of the particle at time $t = 3$ is -1.760.

0	1	2	3

The student earns all of the following point:

1 point is earned for $\int_0^3 v(t)\,dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t)\,dt = -5 + \int_0^3 v(t)\,dt = -1.760213$$

The position of the particle at time $t = 3$ is -1.760.

Part C

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t)\,dt$ and $\int_0^{3.5} |v(t)|\,dt$

$$\int_0^{3.5} v(t)\,dt = 2.844 \text{ (or } 2.843)$$

$$\int_0^{3.5} |v(t)|\,dt = 3.737$$

The integral $\int_0^{3.5} v(t)\,dt$ is the displacement of the particle over the time interval $0 \le t \le 3.5$.

2-6-7

The integral $\int_0^{3.5} |v(t)|\, dt$ is the total distance traveled by the particle over the time interval $0 \le t \le 3.5$.

0	1	2	3

The student earns all of the following point:

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t)\, dt$ and $\int_0^{3.5} |v(t)|\, dt$

$$\int_0^{3.5} v(t)\, dt = 2.844 \text{ (or } 2.843)$$

$$\int_0^{3.5} |v(t)|\, dt = 3.737$$

The integral $\int_0^{3.5} v(t)\, dt$ is the displacement of the particle over the time interval $0 \le t \le 3.5$.

The integral $\int_0^{3.5} |v(t)|\, dt$ is the total distance traveled by the particle over the time interval $0 \le t \le 3.5$.

Part D

1 point is earned if sets $v(t) = {x_2}'(t)$

1 point is earned for the answer

$$v(t) = {x_2}'(t)$$

$v(t) = 2t - 1 \Rightarrow t = 1.57054$

2-6-7

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

0	1	2

The student earns all of the following point:

1 point is earned if sets $v(t) = {x_2}'(t)$

1 point is earned for the answer

$v(t) = {x_2}'(t)$

$v(t) = 2t - 1 \Rightarrow t = 1.57054$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

2-6-7

54. A particle moves along the x-axis with position at time t given by $x(t) = e^{-t}\sin t$ for $0 \leq t \leq 2\pi$.

(a) Find the time t at which the particle is farthest to the left. Justify your answer.

(b) Find the value of the constant A for which $x(t)$ satisfies the equation $Ax''(t) + x'(t) + x(t) = 0$ for $0 < t < 2\pi$.

Part A

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$x'(t) = -e^{-t}\sin t + e^{-t}\cos t = e^{-t}(\cos t - \sin t)$
$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on
$0 \leq t \leq 2\pi$ for $t = \frac{\pi}{4}$ and $t = \frac{5\pi}{4}$.

The candidates for the absolute minimum are at $t = 0, \dfrac{\pi}{4}, \dfrac{5\pi}{4}$, and 2π.

t	$x(t)$
0	$e^0\sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}}\sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}}\sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi}\sin(2\pi) = 0$

The particle is farthest to the left when $t = \dfrac{5\pi}{4}$.

0	1	2	3	4	5

2-6-7

The student response earns all of the following points:

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$x'(t) = -e^{-t}\sin t + e^{-t}\cos t = e^{-t}(\cos t - \sin t)$
$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on
$0 \le t \le 2\pi$ for $t = \frac{\pi}{4}$ and $t = \frac{5\pi}{4}$.

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π.

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

Part B

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

2-6-7

$$x''(t) = -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t)$$
$$= -2e^{-t}\cos t$$
$$Ax''(t) + x'(t) + x(t)$$
$$= A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t$$
$$= (-2A+1)e^{-t}\cos t$$
$$= 0$$

Therefore, $A = \frac{1}{2}$.

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

$$x''(t) = -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t)$$
$$= -2e^{-t}\cos t$$
$$Ax''(t) + x'(t) + x(t)$$
$$= A\left(-2e^{-t}\cos t\right) + e^{-t}(\cos t - \sin t) + e^{-t}\sin t$$
$$= (-2A+1)e^{-t}\cos t$$
$$= 0$$

Therefore, $A = \frac{1}{2}$.

2-6-7

55. Let R be the region bounded by the graphs of $y = 2x$ and $y = 4x - x^2$. What is the area of R ?

(A) $\frac{2}{3}$

(B) $\frac{4}{3}$ ✓

(C) $\frac{16}{3}$

(D) $\frac{28}{3}$

Answer B

Correct. The graphs of $y = 2x$ and $y = 4x - x^2$ intersect when $x = 0$ and $x = 2$. The graph of $y = 4x - x^2$ lies above the graph $y = 2x$ on the interval $0 \leq x \leq 2$. (One way to see this is to sketch a graph of the parabola and the line, observing that the graph of $y = 4x - x^2$ has a slope of 4 at $x = 0$, while the graph of $y = 2x$ has a slope of 2 .) The area of the region bounded by the two graphs is therefore

$$\int_0^2 \left(4x - x^2 - 2x\right) dx = \int_0^2 \left(2x - x^2\right) dx = \left(x^2 - \frac{x^3}{3}\right)\Big|_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

56. $\lim\limits_{h \to 0} \frac{7e^x - 7e^{x+h}}{4h} =$

(A) $-7e^x$

(B) $7e^x$

(C) $-\frac{7}{4}e^x$ ✓

(D) $\frac{7}{4}e^x$

Answer C

Correct.

$$\lim_{h \to 0} \frac{7e^x - 7e^{x+h}}{4h} = \frac{7}{4} \lim_{h \to 0} \frac{e^x - e^{x+h}}{h} = -\frac{7}{4} \lim_{h \to 0} \frac{e^{x+h} - e^x}{h} = -\frac{7}{4}\left(\frac{d}{dx} e^x\right) = -\frac{7}{4} e^x$$

2-6-7

57. $\lim_{h \to 0} \frac{5e^x - 5e^{x+h}}{3h} =$

(A) $-5e^x$

(B) $5e^x$

(C) $-\frac{5}{3}e^x$

(D) $\frac{5}{3}e^x$

Answer C

Correct. $\lim_{h \to 0} \frac{5e^x - 5e^{x+h}}{3h} = \frac{5}{3}\lim_{h \to 0} \frac{e^x - e^{x+h}}{h} = -\frac{5}{3}\lim_{h \to 0} \frac{e^{x+h} - e^x}{h} = -\frac{5}{3}\left(\frac{d}{dx}e^x\right) = -\frac{5}{3}e^x$

58. If $f(x) = \ln x$, then $\lim_{x \to 2} \frac{f(2) - f(x)}{x - 2} =$

(A) $-\ln 2$

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) $\ln 2$

Answer B

Correct. Recognizing that the limit expression is related to the definition of the derivative of $\ln x$ at $x = 2$ offers a strategy for determining the limit.
$\lim_{x \to 2} \frac{\ln 2 - \ln x}{x - 2} = -\lim_{x \to 2} \frac{\ln x - \ln 2}{x - 2} = -\frac{d}{dx}(\ln x)\Big|_{x=2} = -\frac{1}{x}\Big|_{x=2} = -\frac{1}{2}$

2-6-7

59. $\lim_{x \to 0} \frac{2^x - 1}{x} =$

(A) 0

(B) $\ln 2$ ✓

(C) 1

(D) $\frac{1}{\ln 2}$

Answer B

Correct. The limit expression is the definition of the derivative of 2^x at $x = 0$.
$\frac{d}{dx}(2^x)\Big|_{x=0} = (\ln 2 \cdot 2^x)\Big|_{x=0} = \ln 2 \cdot 2^0 = \ln 2$

60. If $f(x) = \sin x$, then $\lim_{x \to 2\pi} \frac{f(2\pi) - f(x)}{x - 2\pi} =$

(A) -2π

(B) -1 ✓

(C) 1

(D) 2π

Answer B

Correct. Recognizing that the limit expression is related to the definition of the derivative of $\sin x$ at $x = 2\pi$ offers a strategy for determining the limit.
$\lim_{x \to 2\pi} \frac{\sin(2\pi) - \sin x}{x - 2\pi} = -\lim_{x \to 2\pi} \frac{\sin x - \sin(2\pi)}{x - 2\pi} = -\frac{d}{dx}(\sin x)\Big|_{x=2\pi} = -\cos x|_{x=2\pi} = -1$

2-6-7

61. $\lim_{x \to 0} \frac{5^x - 1}{x} =$

(A) 0

(B) $\ln 5$ ✓

(C) $5 \ln 5$

(D) 1

Answer B

Correct. The limit expression is the definition of the derivative of 5^x at $x = 0$.
$\frac{d}{dx}(5^x)\Big|_{x=0} = (\ln 5 \cdot 5^x)\Big|_{x=0} = \ln 5 \cdot 5^0 = \ln 5$

62. $\lim_{h \to 0} \frac{\cos\left(\frac{\pi}{5} + h\right) - \cos\left(\frac{\pi}{5}\right)}{h}$ is

(A) $-\sin\left(\frac{\pi}{5}\right)$ ✓

(B) $\sin\left(\frac{\pi}{5}\right)$

(C) $\cos\left(\frac{\pi}{5}\right)$

(D) nonexistent

Answer A

Correct. The derivative of a function f is the function whose value at x is $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.
If $f(x) = \cos x$, then $\lim_{h \to 0} \frac{\cos\left(\frac{\pi}{5} + h\right) - \cos\left(\frac{\pi}{5}\right)}{h}$ is the derivative of $\cos x$ at $x = \frac{\pi}{5}$. Since the derivative of $\cos x$ is $-\sin x$, $\lim_{h \to 0} \frac{\cos\left(\frac{\pi}{5} + h\right) - \cos\left(\frac{\pi}{5}\right)}{h} = -\sin\left(\frac{\pi}{5}\right)$.

2-6-7

63. Let f and g be differentiable functions such that $f'(1) = 2$ and $g'(1) = 6$. If $h(x) = 5f(x) - 4g(x) + 3x^2 - 2$, what is the value of $h'(1)$?

(A) -10

(B) -8

(C) 2

(D) 40

Answer B

Correct. The derivative rules for sums, differences, and constant multiples of functions are applied and correct values are substituted, as follows.
$h'(x) = 5f'(x) - 4g'(x) + 6x$
$h'(1) = 5f'(1) - 4g'(1) + 6 = 5 \cdot 2 - 4 \cdot 6 + 6 = -8$

64. Let f and g be differentiable functions such that $f'(0) = 3$ and $g'(0) = 7$. If $h(x) = 3f(x) - 2g(x) - 5\cos x - 3$, what is the value of $h'(0)$?

(A) -8

(B) -5

(C) 1

(D) 28

Answer B

Correct. The derivative rules for sums, differences, and constant multiples of functions are applied and correct values are substituted, as follows. $h'(x) = 3f'(x) - 2g'(x) + 5\sin x$
$h'(0) = 3 \cdot 3 - 2 \cdot 7 + 5 \cdot 0 = -5$

2-6-7

65. If $f\left(x\right) = 12x^{\frac{1}{2}} + 3x + \frac{32}{x} + 5$, then $f'\left(4\right) =$

(A) 4 ✓

(B) 8

(C) 9

(D) 13

Answer A

Correct. The power rule is used to compute the derivative, as follows.

$f'\left(x\right) = 6x^{-\frac{1}{2}} + 3 - \frac{32}{x^2}$

$f'\left(4\right) = \frac{6}{2} + 3 - \frac{32}{16} = 3 + 3 - 2 = 4$

66. Let f be the function defined by $f\left(x\right) = bx^2 + 3bx + b^2 + \frac{1}{x^2}$, where b is a nonzero constant. Which of the following is an expression for $f'\left(x\right)$, the derivative of $f\left(x\right)$?

(A) $2bx + 3b - \frac{2}{x}$

(B) $2bx + 3b - \frac{2}{x^3}$ ✓

(C) $2bx + 3b + b^2 - \frac{2}{x^3}$

(D) $x^2 + 3x + 2b$

Answer B

Correct. This question involves using the basic power rule for differentiation with respect to the independent variable x while treating the parameter b as a constant, as follows.

$f'(x) = \frac{d}{dx}\left(bx^2 + 3bx + b^2 + x^{-2}\right) = 2bx + 3b + 0 + (-2)x^{-3} = 2bx + 3b - \frac{2}{x^3}$

2-6-7

67. Let f be the function given by $f(x) = 3\sin x + 8e^x$. What is the value of $f'(0)$?

(A) 3

(B) 5

(C) 8

(D) 11 ✓

Answer D

Correct. $f'(x) = 3\cos x + 8e^x$, and therefore $f'(0) = 3\cos 0 + 8e^0 = 3 + 8 = 11$.

68. Let f be the function given by $f(x) = 2\cos x + 3e^x$. What is the value of $f'(0)$?

(A) -3

(B) 0

(C) 3 ✓

(D) 5

Answer C

Correct. $f'(x) = -2\sin x + 3e^x$, and therefore $f'(0) = -2\sin 0 + 3e^0 = 0 + 3 = 3$.

2-6-7

69. Let g be the function given by $g(x) = \lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h}$. What is the instantaneous rate of change of g with respect to x at $x = \frac{\pi}{3}$?

(A) $-\frac{\sqrt{3}}{2}$ ✓

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) $\frac{\sqrt{3}}{2}$

Answer A

Correct. The function g is expressed as the limit of a difference quotient that represents the derivative of $\sin x$. Therefore, $g(x) = \frac{d}{dx} \sin x = \cos x$. The instantaneous rate of change of g with respect to x at $x = \frac{\pi}{3}$ is $g'\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$.

70. Let g be the function given by $g(x) = \lim_{h \to 0} \frac{\cos(x+h) - \cos x}{h}$. What is the instantaneous rate of change of g with respect to x at $x = \frac{\pi}{3}$?

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$ ✓

(D) $-\frac{\sqrt{3}}{2}$

Answer C

Correct. The function g is expressed as the limit of a difference quotient that represents the derivative of $\cos x$. Therefore, $g(x) = \frac{d}{dx} \cos x = -\sin x$. The instantaneous rate of change of g with respect to x is $g'(x) = -\cos(x)$. At $x = \frac{\pi}{3}$, $g'\left(\frac{\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right) = -\frac{1}{2}$.

2-6-7

71. Let g be the function given by $g(x) = x^4 - 2x^3 - 3x$. What are all values of x such that $g'(x) = \frac{1}{2}$?

(A) -4.00

(B) 1.746

(C) 1.777 ✓

(D) -0.164 and 2.508

Answer C

Correct. This question involves using the power rule for differentiation of polynomial functions to find $g'(x) = 4x^3 - 6x^2 - 3$ and then solving an equation. The calculator is used to solve $4x^3 - 6x^2 - 3 = \frac{1}{2}$.

72. Let f be the function given by $f(x) = 2x^3 + x^2 - 3$. What is the value of $f'(2)$?

(A) 56

(B) 28 ✓

(C) 25

(D) 10

Answer B

Correct. This question involves using the power rule for differentiation of polynomial functions and then correctly evaluating at $x = 2$. $f'(x) = 6x^2 + 2x$, so $f'(2) = 6(4) + 2(2) = 28$.

2-6-7

73. Let g be the function given by $g(x) = x^4 - 3x^3 - x$. What are all values of x such that $g'(x) = \frac{1}{2}$?

(A) -2.750

(B) 2.297

(C) 2.320 ✓

(D) -0.353 and 3.119

Answer C

Correct. This question involves using the power rule for differentiation of polynomial functions to find $g'(x) = 4x^3 - 9x^2 - 1$ and then solving an equation. The calculator is used to solve $4x^3 - 9x^2 - 1 = \frac{1}{2}$.

74. Let f be the function given by $f(x) = x^3 + 3x^2 - 4$. What is the value of $f'(2)$?

(A) 48

(B) 24 ✓

(C) 20

(D) 10

Answer B

Correct. This question involves using the power rule for differentiation of polynomial functions and then correctly evaluating at $x = 2$. $f'(x) = 3x^2 + 6x$, so $f'(2) = 12 + 12 = 24$.

2-6-7

75. If $g(x) = 4\cos x + 2\sin x + 1$, then $g'\left(\frac{\pi}{6}\right) =$

(A) $-2 + \sqrt{3}$

(B) $2 - \sqrt{3}$

(C) $2 + \sqrt{3}$

(D) $2 + 2\sqrt{3}$

Answer A

Correct. This question involves using the basic rules for the differentiation of trigonometric functions, and then evaluating the derivative at $\frac{\pi}{6}$.

$g'(x) = -4\sin x + 2\cos x$

$g'\left(\frac{\pi}{6}\right) = -4\sin\left(\frac{\pi}{6}\right) + 2\cos\left(\frac{\pi}{6}\right) = -4 \cdot \frac{1}{2} + 2 \cdot \frac{\sqrt{3}}{2} = -2 + \sqrt{3}$

76. If $g(x) = 3\sin x + 2\cos x + 5$, then $g'\left(\frac{\pi}{3}\right) =$

(A) $\frac{3}{2} - \sqrt{3}$

(B) $-\frac{3}{2} + \sqrt{3}$

(C) $\frac{3}{2} + \sqrt{3}$

(D) $6 + \frac{3}{2}\sqrt{3}$

Answer A

Correct. This question involves using the basic rules for the differentiation of trigonometric functions, and then evaluating the derivative at $\frac{\pi}{3}$.

$g'(x) = 3\cos x - 2\sin x$

$g'\left(\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{3}\right) - 2\sin\left(\frac{\pi}{3}\right) = 3\left(\frac{1}{2}\right) - 2\left(\frac{\sqrt{3}}{2}\right) = \frac{3}{2} - \sqrt{3}$

2-6-7

77. If $f(x) = 4x^6 - 3x^4 + 2x^3 + e^2$, then $f'(x) =$

(A) $4x^5 - 3x^3 + 2x^2$

(B) $24x^5 - 12x^3 + 6x^2$

(C) $24x^5 - 12x^3 + 6x^2 + 2e$

(D) $24x^6 - 12x^4 + 6x^3$

Answer B

Correct. This question involves using the power rule $\frac{d}{dx}x^n = nx^{n-1}$ for differentiation of polynomial functions as follows.
$f'(x) = 4 \cdot 6x^5 - 3 \cdot 4x^3 + 2 \cdot 3x^2 + 0$

78. If $y = \sqrt{x^9}$, then $\frac{dy}{dx} =$

(A) $\frac{9\sqrt{x^7}}{2}$

(B) $3\sqrt{x^8}$

(C) $\frac{1}{2\sqrt{x^9}}$

(D) $\frac{2\sqrt{x^{11}}}{11}$

Answer A

Correct. Using the power rule,
$\frac{d}{dx}\sqrt{x^9} = \frac{d}{dx}x^{\frac{9}{2}} = \frac{9}{2}x^{\frac{7}{2}} = \frac{9\sqrt{x^7}}{2}$.
Alternatively, by the chain rule and power rule,
$\frac{d}{dx}\sqrt{x^9} = \frac{1}{2\sqrt{x^9}} \cdot 9x^8 = \frac{9}{2}x^{8-\frac{9}{2}} = \frac{9}{2}x^{\frac{7}{2}} = \frac{9\sqrt{x^7}}{2}$.

2-6-7

79. Let f be the function given by $f(x) = 2x^3 - 4x^2 - 5$. What is the value of $f'(-1)$?

(A) -14

(B) 6

(C) 9

(D) 14 ✓

Answer D

Correct. This question involves using the power rule for differentiation of polynomial functions and then correctly evaluating at $x = -1$. $f'(x) = 6x^2 - 8x$, so $f'(-1) = 6 + 8 = 14$.

80. If $f(x) = 5x^6 - 3x^5 + 2x^3 - x^2 + e^3$, then $f'(x) =$

(A) $5x^5 - 3x^4 + 2x^2 - x$

(B) $30x^5 - 15x^4 + 6x^2 - 2x$ ✓

(C) $30x^5 - 15x^4 + 6x^2 - 2x + 3e^2$

(D) $30x^6 - 15x^5 + 6x^3 - 2x^2$

Answer B

Correct. This question involves using the power rule $\frac{d}{dx}x^n = nx^{n-1}$ for differentiation of polynomial functions as follows.

$f'(x) = 6 \cdot 5x^5 - 5 \cdot 3x^4 + 3 \cdot 2x^2 - 2x^1$

The derivative of e^3 is 0 because it is a constant.

2-6-7

81. Let f be the function given by $f(x) = 5x^3 - 3x - 7$. What is the value of $f'(-2)$?

(A) -114

(B) 17

(C) 50

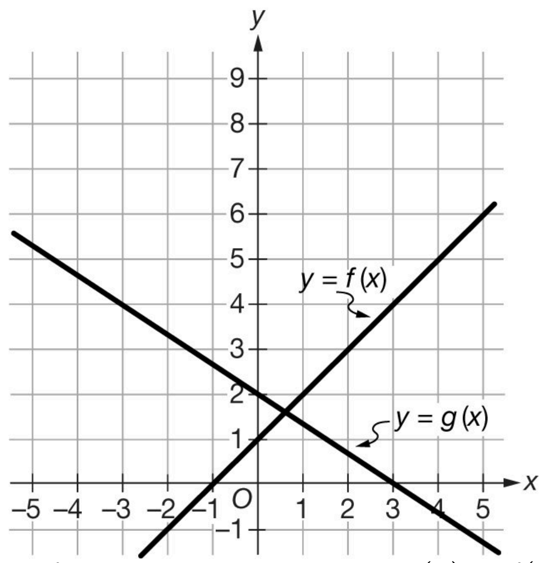
(D) 57

Answer D

Correct. This question involves using the power rule for differentiation of polynomial functions and then correctly evaluating at $x = -2$.
$f'(x) = 15x^2 - 3$, so $f'(-2) = 15 \cdot 4 - 3 = 57$.

2-6-7

82.



The graphs of the linear functions f and g are shown above. If $h(x) = f(x) + g(x)$, then $h'(x) =$

(A) $\frac{5}{3}$

(B) 0

(C) $\frac{1}{3}$

(D) $\frac{1}{3}x + 3$

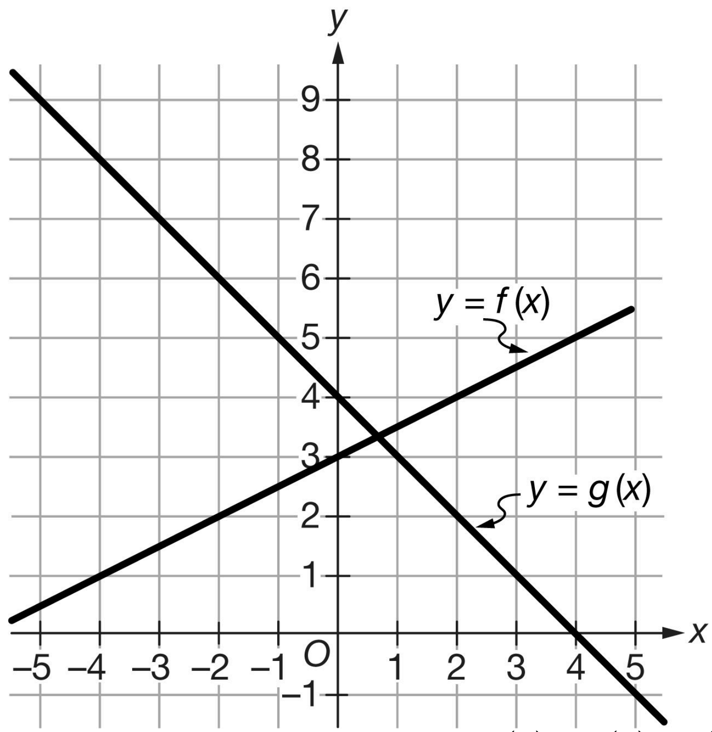
Answer C

Correct. Since f and g are linear functions, their derivatives are equal to their slopes. The derivative of the sum is the sum of the derivatives.

$h'(x) = f'(x) + g'(x) = 1 + \left(-\frac{2}{3}\right) = \frac{1}{3}$

2-6-7

83.



The graphs of the linear functions f and g are shown above. If $h(x) = f(x) + g(x)$, then $h'(x) =$

(A) $\frac{3}{2}$

(B) 0

(C) $-\frac{1}{2}$

(D) $-\frac{1}{2}x + 7$

Answer C

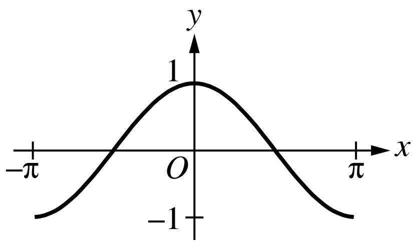
Correct. Since f and g are linear functions, their derivatives are equal to their slopes. The derivative of the sum is the sum of the derivatives.

$h'(x) = f'(x) + g'(x) = \frac{1}{2} + (-1) = -\frac{1}{2}$

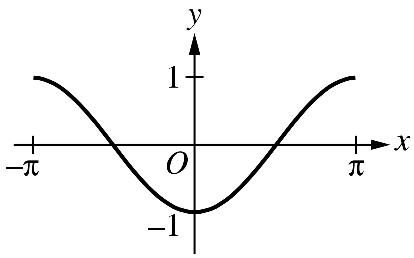
2-6-7

84. Which of the following could be the graph of $y = \frac{\cos(x+0.001)-\cos(x)}{0.001}$?

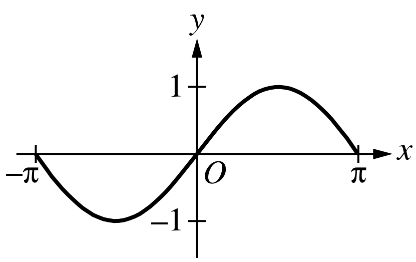
(A)



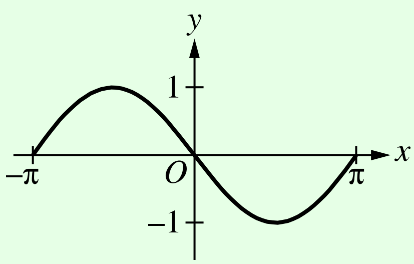
(B)



(C)



(D)



Answer D

Correct. Since $\frac{\cos(x+0.001)-\cos(x)}{0.001} \approx \lim_{h\to 0} \frac{\cos(x+h)-\cos(x)}{h} = \frac{d}{dx}\cos(x) = -\sin(x)$, the graph of $y = \frac{\cos(x+0.001)-\cos(x)}{0.001}$ is closely approximated by the graph of $y = -\sin(x)$. Graph D resembles the graph of $y = -\sin(x)$ on the interval $[-\pi, \pi]$.

2-6-7

85. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant.

(a) Find $f'(x)$ and $f''(x)$.

(b) For what value of the constant k does f have a critical point at $x = 1$? For this value of k, determine whether f has a relative minimum, relative maximum, or neither at $x = 1$. Justify your answer.

(c) For a certain value of the constant k, the graph of f has a point of inflection on the x-axis. Find this value of k.

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$
$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

0	1	2

The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$
$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

Part B

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

2-6-7

$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$

When $k = 2$, $f\prime(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$

When $k = 2$, $f\prime(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

Part C

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

2-6-7

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

Therefore, $\frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$

$\Rightarrow 4 = \ln x$

$\Rightarrow x = e^4$

$\Rightarrow k = \frac{4}{e^2}$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

Therefore, $\frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$

$\Rightarrow 4 = \ln x$

$\Rightarrow x = e^4$

$\Rightarrow k = \frac{4}{e^2}$

2-6-7

86. Let f be the function given by $f(x) = \frac{|x|-2}{x-2}$.

(a) Find all the zeros of f.

(b) Find $f'(1)$.

(c) Find $f'(-1)$.

(d) Find the range of f.

Part A

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2,\ x \neq 2$$
$$x = -2$$

0	1

The student response earns all of the following points:

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2,\ x \neq 2$$
$$x = -2$$

Part B

1 point is earned for $f'(1) = 0$

For $x \geq 0$, $x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$

0	1

The student response earns all of the following points:

2-6-7

1 point is earned for $f'(1) = 0$

For $x \geq 0$, $x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$

Part C

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0,\ f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$
$$f'(x) = \frac{(x-2)(-1)-(-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$
$$\therefore f'(-1) = \frac{4}{9}$$

0	1	2	3	4

The student response earns all of the following points:

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0,\ f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$
$$f'(x) = \frac{(x-2)(-1)-(-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$
$$\therefore f'(-1) = \frac{4}{9}$$

Part D

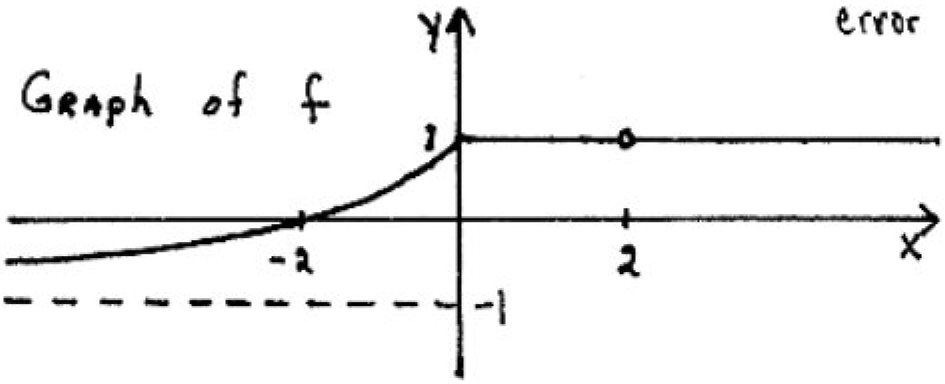
1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$-1 < y \leq 1$

2-6-7



0	1	2	3

The student response earns all of the following points:

1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$-1 < y \leq 1$